

## 1. Measures of Central Tendency

- **AM** =  $\frac{\sum x}{n} \Rightarrow \sum (x - \bar{x}) = 0, \sum (x - \bar{x})^2 = \text{minimum}$  (Extreme values se unaffected)
- **GM** → growth rate, index numbers. **HM** → speed, rate per unit.
- **Median** → positional, extreme values se unaffected, graph = **Ogive**
- **Mode** → highest frequency, graph = **Histogram**, open-ended data ke liye best.

$$\star AM \geq GM \geq HM \quad | \quad GM^2 = AM \times HM \text{ (2 obs.)}$$

$$\star \text{Mode} = 3 \text{ Median} - 2 \text{ Mean} \quad (\text{Empirical Relation})$$

Symmetrical distribution → Mean = Median = Mode

## 2. Measures of Dispersion

| Measure       | Formula                             | Relative Measure                     |
|---------------|-------------------------------------|--------------------------------------|
| Range         | $L - S$                             | $(L - S) / (L + S)$                  |
| Quartile Dev. | $(Q_3 - Q_1) / 2$                   | $(Q_3 - Q_1) / (Q_3 + Q_1)$          |
| Mean Dev.     | $\sum  x - A  / n$                  | MD / A                               |
| Standard Dev. | $\sqrt{[\sum (x - \bar{x})^2 / n]}$ | $CV = (\sigma / \bar{x}) \times 100$ |

- **Kam CV** = zyada consistent / stable — do distributions compare karne ke liye.
- Normal mein → **QD : MD : SD = 10 : 12 : 15** ( $0.6745\sigma, 0.7979\sigma, \sigma$ )
- MD Median se minimum hoti hai. **Lorenz Curve** = graphic method of dispersion.

## 3. Skewness (Asymmetry)

$$\text{Karl Pearson: } Sk = (\text{Mean} - \text{Mode}) / \sigma = 3(\text{Mean} - \text{Median}) / \sigma \quad [-3 \text{ to } +3]$$

$$\text{Bowley: } Sk = (Q_3 + Q_1 - 2M) / (Q_3 - Q_1) \quad [-1 \text{ to } +1, \text{ open-ended ke liye}]$$

| Type          | Relation                           | Tail             |
|---------------|------------------------------------|------------------|
| Symmetrical   | Mean = Median = Mode ; $\mu_3 = 0$ | —                |
| Positive Skew | Mean > Median > Mode               | Right tail lambi |
| Negative Skew | Mean < Median < Mode               | Left tail lambi  |

## 4. Kurtosis (Peakedness)

$$\beta_1 = \mu_3^2 / \mu_2^3 ; \beta_2 = \mu_4 / \mu_2^2 ; \gamma_2 = \beta_2 - 3$$

| Type                | $\beta_2$ | Shape                     |
|---------------------|-----------|---------------------------|
| Leptokurtic         | > 3       | Zyada peaked, patli tails |
| Mesokurtic (Normal) | = 3       | Normal curve              |
| Platykurtic         | < 3       | Flat top                  |

**Trick:** Lepto = Leaping (uchka hua) | Platy = Plateau (flat). Moments:  $\mu_1=0$ ,  $\mu_2=\text{Variance}$ ,  $\mu_3 \rightarrow \text{Skewness}$ ,  $\mu_4 \rightarrow \text{Kurtosis}$

## 5. Elementary Probability

| Approach               | Given by       | Idea                           |
|------------------------|----------------|--------------------------------|
| Classical / Priori     | Laplace        | Favourable / Total             |
| Empirical (Rel. Freq.) | Von Mises      | Large trials ka observed ratio |
| Axiomatic              | Kolmogorov     | Set theory, modern approach    |
| Subjective             | Bayes / Ramsey | Personal belief                |

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) ; P(A/B) = P(A \cap B)/P(B)$$

$$\star \text{ Bayes: } P(A_i/B) = P(A_i) \cdot P(B/A_i) / \sum P(A_i) \cdot P(B/A_i) \rightarrow \text{Prior se Posterior}$$

$$\bullet E(aX+b) = aE(X)+b ; V(aX+b) = a^2V(X) ; V(X)=E(X^2)-[E(X)]^2$$

## 6. Probability Distributions

| Basis           | Binomial (Bernoulli)          | Poisson                       | Normal (Gauss)         |
|-----------------|-------------------------------|-------------------------------|------------------------|
| Formula         | ${}^nC_r p^r q^{n-r}$         | $e^{-\lambda} \lambda^r / r!$ | $Z = (X-\mu)/\sigma$   |
| Parameters      | $n, p$                        | $\lambda$                     | $\mu, \sigma$          |
| Mean            | $np$                          | $\lambda$                     | $\mu$                  |
| Variance        | $npq$                         | $\lambda$                     | $\sigma^2$             |
| Pehchan         | Mean > Variance               | Mean = Variance               | Mean=Median=Mode       |
| Skew / Kurtosis | $p=0.5 \rightarrow$ symmetric | Always +ve skew, Leptokurtic  | $\beta_1=0, \beta_2=3$ |

- Poisson = Binomial ka limiting case ( $n \rightarrow \infty, p \rightarrow 0, np = \lambda$ ). Rare events: accidents, printing mistakes.
- Normal curve x-axis ke liye asymptotic, total area = 1, QD = 0.6745 $\sigma$ , MD = 0.7979 $\sigma$ .

$$\text{Normal Area (ratio): } \mu \pm 1\sigma = 68.27\% \mid \mu \pm 2\sigma = 95.45\% \mid \mu \pm 3\sigma = 99.73\% \mid \mu \pm 1.96\sigma = 95\% \mid \mu \pm 2.58\sigma = 99\%$$

## 7. Correlation

$$r = \text{Cov}(x,y) / (\sigma_x \sigma_y) \quad [\text{Karl Pearson, range } -1 \text{ to } +1]$$

- Origin AUR scale dono se independent, pure number (unit-free),  $r_{xy} = r_{yx}$ .
- $r^2$  = Coefficient of Determination (explained variation) ;  $\sqrt{1-r^2}$  = Alienation.
- PE =  $0.6745(1-r^2)/\sqrt{n} \rightarrow$  agar  $r > 6 \text{ PE}$  to correlation significant.
- Spearman Rank:  $R = 1 - 6\sum D^2 / n(n^2-1) \rightarrow$  qualitative data ke liye.
- Correlation  $\neq$  Causation (Spurious correlation).

## 8. Regression — (term: Sir Francis Galton)

$$b_{yx} = r\sigma_y/\sigma_x ; b_{xy} = r\sigma_x/\sigma_y \quad \star r = \pm\sqrt{(b_{yx} \times b_{xy})}$$

- Dono coefficients ka sign same = r ka sign. Dono 1 se bade nahi ho sakte.
- Origin se independent, par SCALE pe dependent (correlation se ulta!).
- Dono regression lines ( $\hat{x}, \hat{y}$ ) point pe intersect karti hai.
- $r = \pm 1 \rightarrow$  lines coincide  $\mid r = 0 \rightarrow$  lines perpendicular ( $90^\circ$ )
- Standard Error of Estimate =  $\sigma_y\sqrt{1-r^2}$  ; Method = Least Squares [ $\sum (Y-\hat{Y})^2 = \min$ ].

## 9. Testing of Hypothesis (Neyman & Pearson)

|              | $H_0$ Actually TRUE                         | $H_0$ Actually FALSE                        |
|--------------|---------------------------------------------|---------------------------------------------|
| Reject $H_0$ | Type I Error ( $\alpha$ ) — Producer's risk | Correct decision                            |
| Accept $H_0$ | Correct decision                            | Type II Error ( $\beta$ ) — Consumer's risk |

- Power of Test =  $1 - \beta$  ;  $\alpha$  aur  $\beta$  inversely related ; Type I zyada serious.
- Z (two-tailed): 5% =  $\pm 1.96$ , 1% =  $\pm 2.58$  ; p-value <  $\alpha \rightarrow$  Reject  $H_0$
- SE of mean =  $\sigma/\sqrt{n}$  ; SE of proportion =  $\sqrt{PQ/n}$  ; **SE  $\propto 1/\sqrt{n}$**
- CLT:  $n \geq 30$  par sampling distribution Normal ho jati hai.
- Good estimator (Fisher): Unbiasedness, Consistency, Efficiency, Sufficiency.

Tests — Kaun sa, Kisne, Kis liye

| Test       | Given by                | Use                                        | df                             |
|------------|-------------------------|--------------------------------------------|--------------------------------|
| Z-test     | —                       | Large sample ( $n > 30$ ), mean/proportion | —                              |
| t-test     | W.S. Gosset ("Student") | Small sample mean, paired data             | $n-1$ ; $(n_1+n_2-2)$          |
| Chi-square | Karl Pearson            | Goodness of fit, Independence, Homogeneity | <b><math>(r-1)(c-1)</math></b> |
| F / ANOVA  | R.A. Fisher             | Variance equality, 3+ means                | $(n_1-1, n_2-1)$               |

$\chi^2 = \sum(O-E)^2/E \rightarrow$  non-parametric, hamesha +ve & positively skewed. Yates' correction 2x2 table ke liye (freq < 5).  **$t^2 = F$**  (jab df=1). ANOVA: Total = Between (SSB) + Within (SSW), assumption = Homoscedasticity.

## 10. Index Numbers — "Economic Barometers"

| Index          | Formula                                  | Weight / Note                      |
|----------------|------------------------------------------|------------------------------------|
| Laspeyre's     | $\sum p_1 q_0 / \sum p_0 q_0 \times 100$ | Base year $q_0$ — Upward bias      |
| Paasche's      | $\sum p_1 q_1 / \sum p_0 q_1 \times 100$ | Current year $q_1$ — Downward bias |
| Fisher's Ideal | $\sqrt{(L \times P)}$                    | GM of L & P                        |
| Dorbish-Bowley | $(L + P)/2$                              | AM of L & P                        |

- Time Reversal Test:  $P_{01} \times P_{10} = 1$  | Factor Reversal:  $P_{01} \times Q_{01} = \sum p_1 q_1 / \sum p_0 q_0$
- Fisher = "Ideal"  $\rightarrow$  TRT & FRT dono pass, par Circular Test FAIL | Marshall-Edgeworth  $\rightarrow$  sirf TRT pass.
- Circular test pass  $\rightarrow$  Kelly's fixed weight, Simple Aggregative, Simple GM.
- CPI = Laspeyre's type. India: CPI base 2012 (NSO), WPI base 2011-12; RBI target 4%  $\pm$  2%.

## 11. Time Series

Multiplicative (most used):  $Y = T \times S \times C \times I$  | Additive:  $Y = T + S + C + I$

- Trend (T) long-term | Seasonal (S) = 1 saal ke ANDAR | Cyclical (C) = 1 saal se ZYADA | Irregular (I) = random.
- Trend methods: Free-hand, Semi-average, Moving average, **Least Squares (best)**.
- Seasonal measure: Ratio to Moving Average (best), Ratio to Trend, Link Relative, Seasonal Index avg = 100.

## 12. Sampling & Census

| Basis              | Census           | Sample                      |
|--------------------|------------------|-----------------------------|
| Coverage           | Poori population | Population ka part          |
| Sampling Error     | <b>Nahi hota</b> | Hota hai (n badhne par KAM) |
| Non-Sampling Error | <b>Zyada</b>     | Kam                         |

- Probability: Simple Random (lottery / **Tippett's table**), Stratified, Systematic (har  $k^{th}$  unit), Cluster, Multi-stage.
- Non-Probability: Judgement/Purposive, Quota, Convenience, Snowball.
- Stratified  $\rightarrow$  strata andar se HOMOGENEOUS** | **Cluster  $\rightarrow$  cluster andar se HETEROGENEOUS**
- Bias = systematic one-directional error — averaging se khatm nahi hota.

★ Last-minute 8: (1)  $AM \geq GM \geq HM$  (2) Mode =  $3Med - 2Mean$  (3) +ve skew  $\rightarrow Mean > Med > Mode$  (4) Binomial Mean > Var, Poisson Mean = Var (5)  $r = \sqrt{(b_{yx} \times b_{xy})}$  (6) Type I =  $\alpha$  = Producer's risk (7) Laspeyre =  $q_0$ , Paasche =  $q_1$ , Fisher =  $\sqrt{LP}$  (8)  $t \rightarrow$  Gosset,  $\chi^2 \rightarrow$  K. Pearson, F  $\rightarrow$  Fisher